Programming Assignment: "Car Park Management Sim"

Assessment Criteria

1. Software Engineering Quality (10%)

- Clean, modular design; functions/classes with single responsibility
- Encapsulation & Fields (2 pts): private fields, meaningful names
- Constructors & Validation (3 pts): correct defaults
- Getters/Setters & Utility Methods (3 pts): meaningful names
- Style & Documentation (2 pts): Javadoc, formatting, conventions

2. Functional Correctness & Robustness (40%)

3. Thread Safe (15%)

4. Synchronization Implementation (15%):

5. Responsible Generative AI Use & Reflection (20%)

Required Tasks:

Create a Java (JavaFX/Swing) GUI-based simulation called "Car Park Management Sim" to simulate automated car park management. The goal is to keep the car park running without the parking lot (the buffer) overflowing or the car (Consumers) sitting idle at the gate.

- **The Core Entities**
    a. Producers (Generate cars): These generate "Cars" at varying speeds to simulate the car owner is looking for parking lots
    b. The buffer (Parking lots): A fixed-size parking lot. You can set the default parking lot.

c. Consumers (Can be a sign with direction or security guard): to guide the car to the available parking lot

- **Technical Mechanics**
  a. **Multi-Threading & Synchronization**
  b. Each Producer and Consumer should run on its own **Thread**. You must handle the "Critical Section" (the Buffer) using:
  c. **Peterson Solution/Mutexes/Semaphores:** To prevent two threads from grabbing the same item.
  d. **Condition Variables:** To signal Producers to "Sleep" when the parking lot is full and "Wake up" when there's available parking space.
- **Game Variables:**
  a. Allow the user to adjust these in real-time to see how the system reacts:
  b. **Buffer Capacity**
  c. **Production Rate**
  d. **Consumption Rate**
  e. **Processing Time:** Simulates "Work" by putting threads to sleep for milliseconds.
- **Visual Dashboard**
  a. Your UI should display real-time statistics so the user can "see" the code working:
  b. **Occupied Parking Lot(%), Available Parking Lot**, **Throughput** (Items processed per second), **Wait Time** (How long, on average, a Consumer waits for an item), **Thread Status** (Visual icons showing if a thread is **Active**, **Waiting (Blocked)**, or **Crashed)**.

Submission Requirements

- **Complete Simulation Code.**
- **Slide presentation & Presentation Video**
- **AI Appendix:**
    - Tools used (e.g., ChatGPT, Gemini, Copilot) + when/how they were used.
    - Prompts (or paraphrases) and snippets accepted or rejected.
    - Brief reflection: what AI got wrong, how the student verified/modified it, and what they learned.
- Put three requirements into one folder and zip it. Then, rename the zip file to your name and submit